Networks and Complexity

Solution 15-5

This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at https://github.com/NC-Book/NCB

Ex 15.5: Competing epidemics [Lvl. 3]

Consider two competing SIS diseases I and J, which spread at different rates $p_1 \neq p_2$ but from which people recover at the same rate r. Assume that being infected with one of the diseases gives the infected person immunity to the other disease. Use a mean field approximation to show that the system cannot be in a stationary state where both diseases persist.

Solution

Let us call the diseases I and J. We write the mean field equations for the proportion of nodes that are infected with I or J, respectively.

$$[\dot{I}] = p_1[I][S] - r[I] \tag{1}$$
$$[\dot{J}] = p_2[J][S] - r[J] \tag{2}$$

where $[S] = 1 - [I] - [J]$. A steady state needs to obey the conditions

$$0 = p_1[I][S] - r[I] \tag{3}$$
$$0 = p_2[J][S] - r[J] \tag{4}$$

and because we are interested in nonzero solutions we can divide by $[I]$ and $[J]$, respectively, so

$$0 = p_1[S] - r \tag{5}$$
$$0 = p_2[S] - r. \tag{6}$$

Now we can see that there is no value of $[S]$ where both of these conditions can be met simultaneously, unless $p_1 = p_2$. Hence for $p_1 \neq p_2$ there isn't a steady state where both diseases coexist.

This outcome is an example for the competitive-exclusion principle. The number of species in a system cannot be greater than the number of niches.